CPS Smart Grid Simulation Report
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Introduction
This report reproduces the course project pipeline for the custom 5-bus DC network:
DC power flow, PTDF, LODF, DCOPF, N-1 screening, single-contingency SCOPF, and the
operating-state comparison.
Use of generative AI tools is disclosed for implementation assistance. Numerical values
are produced by `scripts/run_project.py`.
Phase 1: Ybus/DC PF/PTDF
Base dispatch for the foundation check: `{1: 110.0, 2: 40.0, 3: 0.0}` MW.
Angles in radians:

Bus	theta
1	0.0440
2	-0.0288
3	0.0176
4	0.0000
5	-0.0503

Line-flow cross-check:

Line	Direct DC PF MW	PTDF MW
L1	77.4161	77.4161
L2	32.5839	32.5839
L3	-25.6049	-25.6049
L4	14.3951	14.3951
L5	46.9790	46.9790
L6	13.0210	13.0210

Maximum cross-check error: `0.00000000` MW.
Slack column of PTDF is zero: `True`.
PTDF:
| | Bus 1 | Bus 2 | Bus 3 | Bus 4 | Bus 5 |

L1	0.3626	-0.3850	-0.1550	0.0000	-0.1514
L2	0.6374	0.3850	0.1550	0.0000	0.1514
L3	0.1715	0.2909	-0.2855	0.0000	0.1145
L4	0.1715	0.2909	0.7145	0.0000	0.1145
L5	-0.1911	-0.3241	-0.1305	0.0000	-0.7341
L6	0.1911	0.3241	0.1305	0.0000	-0.2659

Phase 2: DCOPF And LMPs

Base DCOPF cost: `1860.00` USD/h.
Base dispatch: `{1: 150.0, 2: 0.0, 3: 0.0}` MW.
Base LMPs: `['12.40', '12.40', '12.40', '12.40', '12.40']` USD/MWh.
Binding base lines: `[]`.
With engineered congestion `L2 = 30 MW`, cost is
`2322.63` USD/h and dispatch is
`{1: 104.64388760868304, 2: 45.35611239131693, 3: 0.0}` MW.

Phase 3: LODF And N-1 Screening

LODF matrix:

	L1	L2	L3	L4	L5	L6
L1	1.0000	1.0000	-0.5429	-0.5429	0.5695	-0.5695
L2	1.0000	1.0000	0.5429	0.5429	-0.5695	0.5695
L3	-0.4730	0.4730	1.0000	-1.0000	-0.4305	0.4305
L4	-0.4730	0.4730	-1.0000	1.0000	-0.4305	0.4305
L5	0.5270	-0.5270	-0.4571	-0.4571	1.0000	1.0000
L6	-0.5270	0.5270	0.4571	0.4571	1.0000	1.0000

Explicit LODF value `L_L3,L1 = -0.4730`.
Contingency table:

Outage	Max loading %	Worst line	Overloaded
L1	150.00	L2	True
L2	150.00	L1	True
L3	102.10	L1	True
L4	102.10	L1	True
L5	123.50	L1	True

L6	89.32	L1	False

Worst contingency: `L2`.
Physical/adversarial event: A plausible event is a relay misconfiguration or breaker operation that trips L2 durin

Phase 4: SCOPF

SCOPF enforced contingencies: `[2]`.
SCOPF cost: `2370.00` USD/h.
SCOPF dispatch: `{1: 99.99999999999999, 2: 50.000000000000014, 3: 0.0}` MW.
SCOPF LMPs: `['12.40', '22.60', '22.60', '22.60', '22.60']` USD/MWh.
Dispatch comparison:

Generator	Base DCOPF MW	SCOPF MW	Move MW
G1	150.0000	100.0000	-50.0000
G2	0.0000	50.0000	50.0000
G3	0.0000	0.0000	0.0000

Cost of security: `510.00` USD/h
(`27.42%` of base DCOPF cost).
SCOPF post-contingency max loading under the selected outage:
`100.00%`.

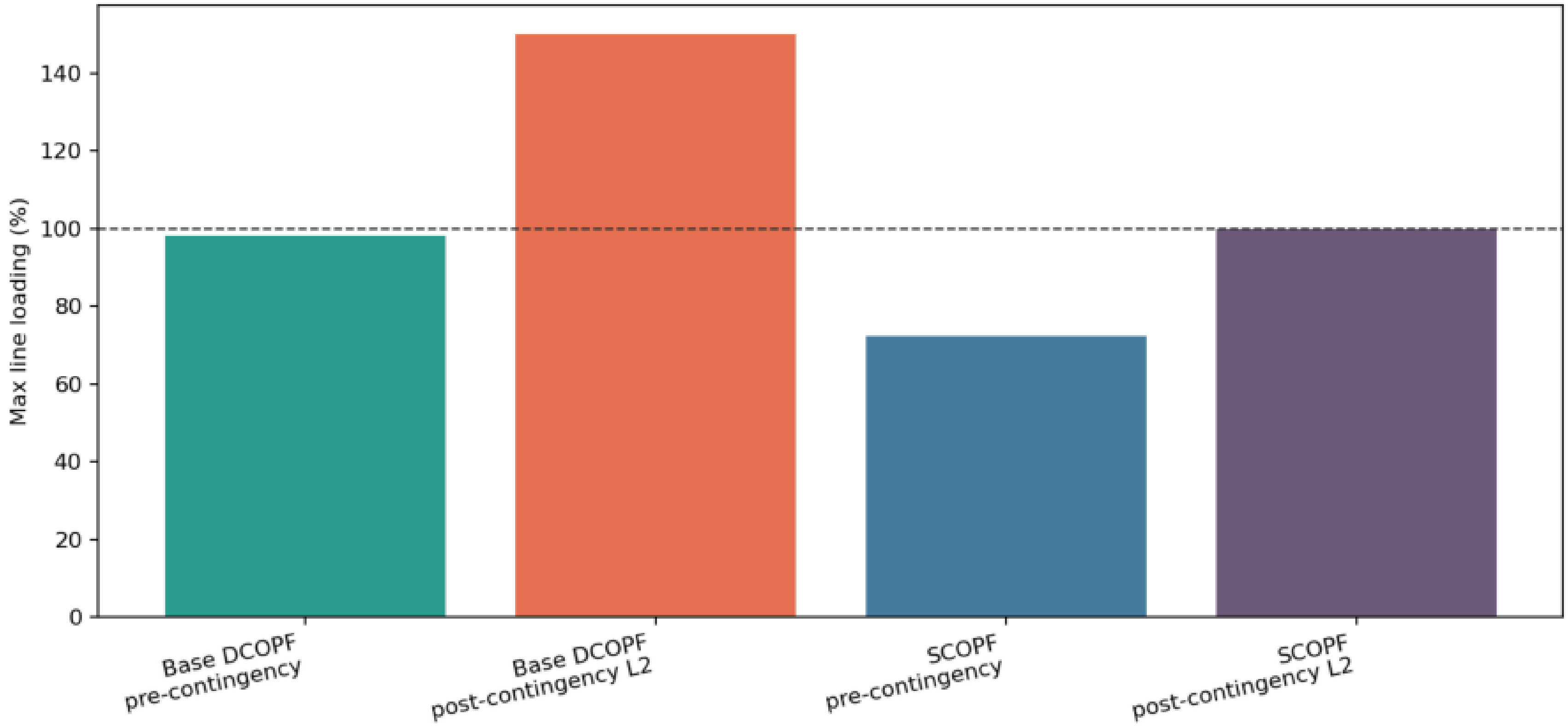
Phase 5: Operating-State Analysis

Dispatch	Scenario	Max loading %	Cost USD/h
Base DCOPF	pre-contingency	98.12	1860.00
Base DCOPF	post-contingency L2	150.00	1860.00
SCOPF	pre-contingency	72.24	2370.00
SCOPF	post-contingency L2	100.00	2370.00

Interpretation: SCOPF pays the preventive cost of security every hour, but removes the overload exposure for the s
Cyber-physical reflection: The Ukraine 2015 power-grid attack showed that adversaries can use cyber access to crea

Conclusion

The implementation provides a reproducible numerical pipeline and an API surface for
rerunning the project calculations locally or in Docker.


Max line loading (%)
0
20
40
60
80
100
120
140
Base DCOPF
pre-contingency
Base DCOPF
post-contingency L2
SCOPF
pre-contingency
SCOPF
post-contingency L2